

Nasif Fahmid Prangon

PhD Researcher, Computer and Information Sciences
Temple University, USA

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Research Interests

Edge Intelligence, Distributed Learning Systems, Knowledge Distillation, Federated Learning, Split Learning, Network-Aware Machine Learning, Cloud and Edge Computing.

Education

Temple University, USA

Ph.D. in Computer and Information Sciences

Advisor: Dr. Jie Wu

Aug 2023 – Present

CGPA: 3.83/4.00

Rajshahi University of Engineering and Technology (RUET), Bangladesh

B.Sc. in Electronics and Telecommunication Engineering

2014 – 2018

CGPA: 3.44/4.00

Publications

ToCo-Split: Topology-Constrained Communication for Efficient Split Learning in Edge Intelligence

N. F. Prangon and J. Wu

23rd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), 2026.

Key Idea: Topology-aware split learning combining cut-layer selection, distribution-aware client clustering, and structured gradient mixing for communication-efficient edge intelligence.

FedBaS: Bandwidth-Aware Feature Selection Strategies for Efficient Federated Distillation

N. F. Prangon and J. Wu

2026 IEEE 104th Vehicular Technology Conference (VTC2026-Fall), 2026.

Key Idea: Bandwidth-aware federated distillation using selective feature sharing to improve the communication-accuracy trade-off in resource-constrained edge networks.

Beyond the Teacher-Student Paradigm: The Role of Network Structure in Knowledge Distillation

N. F. Prangon and J. Wu

International Conference on Computing, Networking and Communications (ICNC), 2026.

Key Idea: Topology-aware knowledge distillation showing how structures like hypercube, ring, and chain impact learning efficiency in distributed systems.

Resource Optimized Task Offloading in Delay Sensitive Edge Networks

N. F. Prangon, A. Sawwan, and J. Wu

International Conference on Computing, Networking and Communications (ICNC), 2025.

Key Idea: Energy- and delay-aware task allocation for edge networks.

AI and Computing Horizons: Cloud and Edge in the Modern Era

N. F. Prangon and J. Wu

Journal of Sensor and Actuator Networks (JSAN), 2024.

Key Idea: Survey of distributed AI and cloud-edge intelligence.

Research Experience

Graduate Research Assistant

Aug 2023 – Present

Temple University

- Conducting research on the impact of topology and network structure for distributed edge AI systems.
- Designing communication-efficient learning frameworks using structured network topologies.
- Working on LLM applications in healthcare (Summer 2024, Summer 2025).

Teaching Experience

Graduate Teaching Assistant

Temple University

Aug 2023 – Present

- Computer Systems and Low-Level Programming (C, memory, pointers, OS concepts)

Industry Experience

Key Account Manager (Wholesale)

Axiata Group, Malaysia

Oct 2022 – Jul 2023

- Conducted large-scale traffic analysis and capacity planning across international telecom networks, collaborating with multiple OpCos to forecast demand and optimize service delivery.
- Managed roaming and interconnect services (IPX, SCCP, SIGOS), ensuring reliability and performance while evaluating cost-performance trade-offs for efficient resource allocation.

Specialist, Roaming Network (Core Network Planning)

Robi Axiata Limited

Aug 2022 – Jul 2023

- Designed and evolved scalable roaming network architectures, translating business requirements into deployable technical solutions through vendor evaluation and system integration.
- Led capacity planning and network expansion initiatives, coordinating cross-functional teams to maintain performance, reliability, and service quality under increasing traffic demands.

System Engineer (Core Network Operations)

Novotel Ltd

Feb 2020 – Jul 2022

- Managed telecom core infrastructure including SIP signaling, SBCs, and SMS gateways, ensuring high availability through monitoring, troubleshooting, and performance optimization.
- Implemented routing optimizations and maintained carrier integrations (SMPP, HTTP, SS7), leveraging tools such as Wireshark, tcpdump, MySQL, and Elasticsearch for traffic analysis and issue diagnosis.

Technical Skills

Programming: Python, C, SQL, MATLAB

Systems & Tools: Linux, Wireshark, tcpdump, MySQL, Elasticsearch, Kibana

Networking: SIP, SMPP, IP Routing, Core Network Systems

ML/AI: PyTorch, Scikit-learn, Transformers, Time Series Models

Selected Projects

Twitch Friendship Prediction – Graph-based link prediction using graph metrics, embeddings, and ML models.
GitHub

MindGuide – AI-assisted platform for analyzing online mental health resources.

GitHub

EduQuery – Intelligent academic information retrieval system.

GitHub

Languages

English (Professional), Bangla (Native)